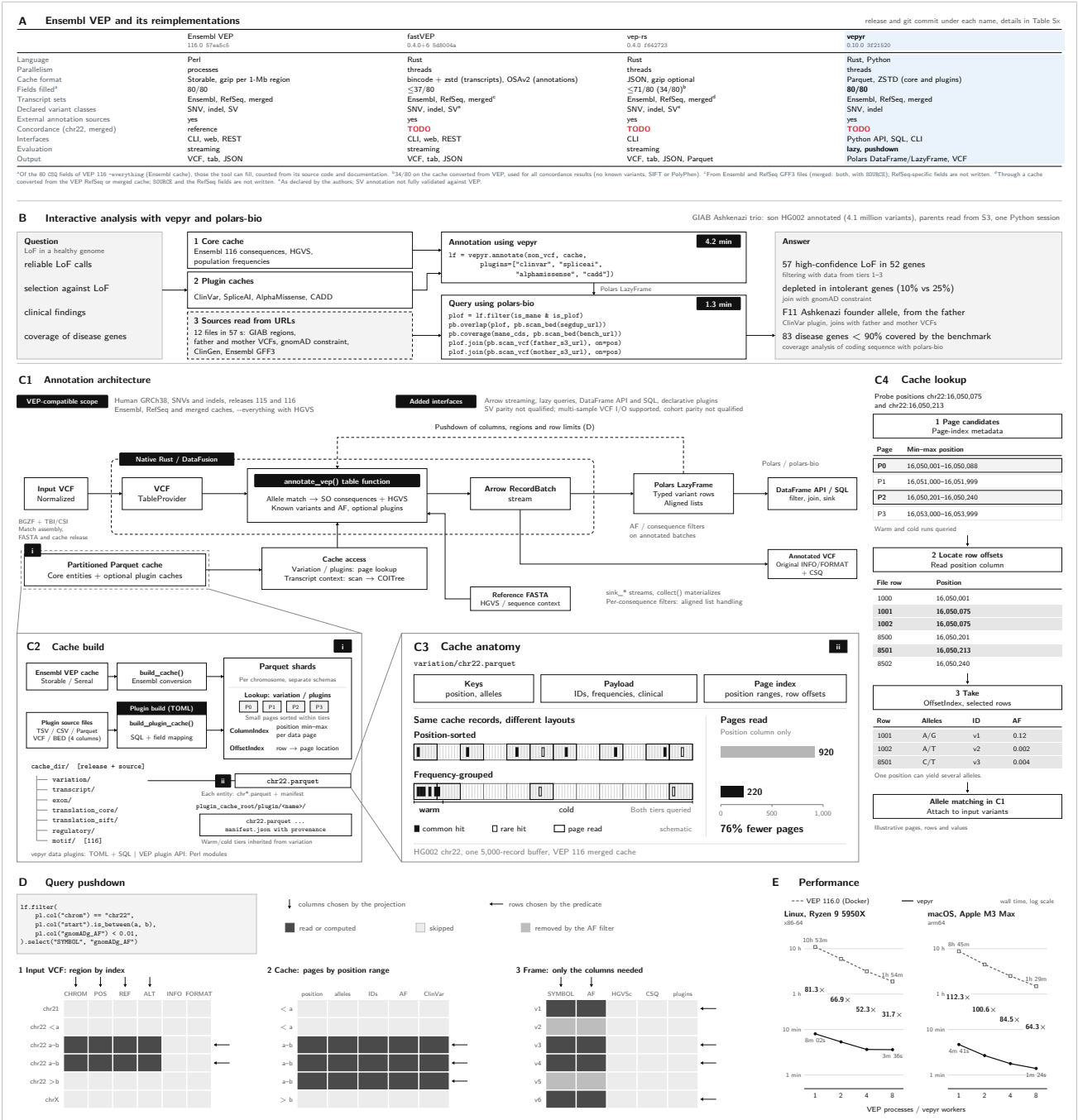


Fig. 1. vepyr: VEP-compatible annotation as a query in the Python DataFrame ecosystem. (A) Ensembl VEP and three reimplementations. *Fields filled*: of the 80 `CSQ` fields of VEP 116 `--everything` (Ensembl cache), those each tool can fill. *Concordance*: on chromosome 22 with the merged cache, same inputs for all tools [TODO]. **(B)** The GIAB Ashkenazi trio in one Python session: annotation of the son's genome (HG002, 4.1 million variants) with the core and plugin caches (tiers 1 and 2) took 4.2 min with 8 workers, and queries with polars-bio, including 12 files read from their URLs (tier 3) among them the father's and mother's VCFs, 1.3 min. Of 158 variants with a predicted loss-of-function (LoF) consequence in a MANE Select transcript, 57 in 52 genes pass filters from all three tiers; they are depleted from LoF-intolerant genes (10% in gnomAD LOEUF deciles 1–2 against 25% of rare synonymous variants, $p = 3 \times 10^{-4}$). The only ClinVar pathogenic allele is *F11* Glu117Ter, the Ashkenazi founder allele of factor XI deficiency, inherited from the father; 83 of 2,302 ClinGen disease genes are less than 90% covered by the benchmark regions. **(C1)** Dataflow from the input VCF through the Rust/DataFusion engine to Arrow batches, an annotated VCF or a Polars LazyFrame. **(C2)** Core and plugin caches are built as chromosome-partitioned Parquet with page indexes. **(C3)** Warm and cold frequency tiers reduce the pages read for one HG002 buffer by 76%. **(C4)** Lookup: page index, position column, payload of the matched rows. **(D)** Query pushdown: a genomic predicate selects the VCF region through its index and the cache pages through the page index; the selected and filtered columns decide which annotation is computed; other predicates filter the annotated rows. **(E)** Whole-genome wall time of VEP 116 and vepyr with 1–8 processes or workers (HG002, merged cache, `--everything`).

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